F78PA Probability & Statistics A, December 2023/24 Solutions

Maps to CS1 Actuarial Statistics exemption.
Weighting within CS1 exemption: 25%.

1. *Parts (a) to (c) are similar to seen examples; part (d) unseen.*
CS1 syllabus objective 1.2.1.

(a) Sample size $n = 15$, so

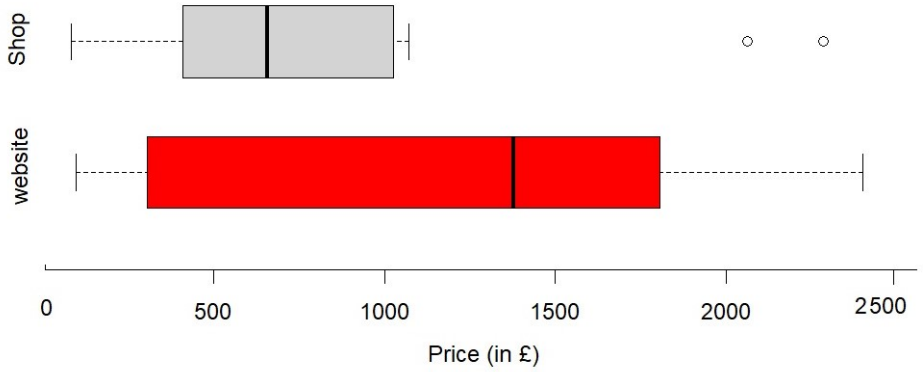
mean, $\bar{x} = \frac{\sum x_i}{n} = \frac{12{,}075}{15} = 805$ (£) [1 mark]

standard deviation, $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{5{,}779{,}310}{14}} \approx 642.501$ (£) [1 mark]

(b)

$$\begin{aligned} \text{median} &= x_{\left(\frac{15+1}{2}\right)} = x_{(8)} = 650 \text{ (£)} && \text{[1 mark]} \\ \text{lower quartile, } Q_1 &= x_{(1+0.25\times(15-1))} = x_{(4.5)} = \frac{350+475}{2} = 412.5 \text{ (£)} \\ \text{upper quartile, } Q_3 &= x_{(1+0.75\times(14-1))} = x_{(11.5)} = \frac{999+1015}{2} = 1007 \text{ (£)} \\ \text{inter-quartile range, } IQR &= 1007 - 412.5 = 594.5 \text{ (£)} && \text{[2 marks]} \end{aligned}$$

(c) To identify any outliers for the sales in the shop,
$Q_1 - 1.5 \times IQR = -479.25$
$Q_3 + 1.5 \times IQR = 1898.75$ $\Rightarrow$ There are two outliers (2100 and 2315) on the right.



[2 marks for boxplots with two outliers using the same scale]

Comments: The distribution of sales in the shop is right-skewed with two outliers of £2100 and £2315, while sales on the website appear to be slightly

left-skewed. The median of the sales on the website is much higher than the median of the sales in the shop (£1380 compared to £650). The ranges of these sales are approximately equal. The IQR of website sales is larger than the IQR of shop sales (£1500 compared to £594.5), and so on.

[2 marks for any two sensible comments]

(d) When data sets are skewed, a suitable numerical measure of spread for comparing their dispersion is the interquartile range (IQR).

The IQR measures the range within which the middle 50% of the data falls, making it robust to outliers and less affected by the skewness of the data compared to standard deviation or variance. It provides a good representation of the spread of the data, especially when dealing with non-normal distributions or data with outliers. [1 mark for sensible choice and comment]

[Total 10 Marks]

2. *Parts (a) to (b) are similar to seen examples; part (c) unseen.*
CS1 syllabus objective 2.1.3.

(a) $f_X(x)$ is a density because $f_X(x) \geq 0$ and

$$\begin{aligned}\int_{x=0}^{10} f(x)\,dx &= \int_{x=0}^{5} \frac{2}{75}x\,dx + \int_{x=5}^{10} \frac{2}{15}\,dx \\ &= \left[\frac{x^2}{75}\right]_{x=0}^{5} + \left[\frac{2x}{15}\right]_{x=5}^{10} \\ &= \left(\frac{1}{3} + \frac{2}{3}\right) \\ &= 1\end{aligned}$$

[1 mark]

(b) $ImX = [0,\, 10]$
So for $x < 0$, $F_X(x) = 0$, and for $x > 10$, $F_X(x) = 1$.
There are two more cases to consider:
For $0 \leq x \leq 5$:

$$F_X(x) = \int_{u=0}^{x} \frac{2}{75}u\,du = \left[\frac{u^2}{75}\right]_{u=0}^{x} = \frac{x^2}{75}$$

For $5 < x \leq 10$:

$$F_X(x) = \frac{5^2}{75} + \int_{u=5}^{x} \frac{2}{15}\,du = \frac{1}{3} + \left[\frac{2}{15}u\right]_{u=5}^{x} = \frac{1}{3} + \frac{2}{15}(x-5) = \frac{2}{15}x - \frac{1}{3}$$

Putting it all together,

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{75}x^2 & 0 \leq x \leq 5, \\ \frac{2}{15}x - \frac{1}{3} & 5 < x \leq 10, \\ 1 & x > 10. \end{cases}$$

[3 marks]

i. $P(X > 4) = 1 - F_X(4) = 1 - \frac{1}{75} \cdot (4^2) = \frac{59}{75} \approx 0.7867$

[1 mark]

ii. Since $F_X(m) = 0.5$, m falls within the third piece-wise segment of $F_X(x)$.

$$\left(\frac{2}{15}m - \frac{1}{3}\right) = 0.5$$

Solve for the above equation, $m = 6.25$. [1 mark]

(c) Let Y be the number of batteries that work within 40 hours in the said electronic device. Using the result from part b(i) above, $Y \sim \text{Bin}(2, \frac{16}{75})$.

$$\begin{aligned} P(\text{device stops working within 40 hours}) &= P(Y \geq 1) = 1 - P(Y = 0) \\ &= 1 - \binom{2}{0}\left(\frac{16}{75}\right)^0\left(\frac{59}{75}\right)^2 \\ &= \frac{2144}{5625} \approx 0.38115 \end{aligned}$$

[2 marks]

[Total 8 Marks]

3. *Most parts (a) seen; parts a(iii) and (b) unseen.*
CS1 syllabus objectives 2.1.2, 2.1.3, 2.1.5.

(a) i. For $w \leq 1$, density is zero, and so $F_W(w) = 0$ for $w \leq 1$.
For For $w > 1$,

$$\begin{aligned} F_W(w) = P(W \leq w) &= \int_1^w f_W(u)\,du = \int_1^w u^{-2}\,du \\ &= \left[-\frac{1}{u}\right]_1^w \\ &= 1 - \frac{1}{w} \end{aligned}$$

That is,

$$F_W(w) = \begin{cases} 0 & w \leq 1, \\ 1 - \frac{1}{w} & w > 1. \end{cases}$$

[2 marks]

ii. Given that $T = \min\{W_1, W_2, \ldots, W_n\}$ and $ImW = (1, \infty)$, so $ImT = (1, \infty)$. For $t \leq 1$, density is zero, and so $F_T(t) = 0$. For $t > 1$,

$$\begin{aligned} F_T(t) = P(T \leq t) &= P(\min\{W_1, W_2, \ldots, W_n\} \leq t) \\ &= 1 - P(\min\{W_1, W_2, \ldots, W_n\} > t) \\ &= 1 - P(W_1 > t, W_2 > t, \ldots, W_n > t) \\ &= 1 - P(W_1 > t)^n \qquad \text{(Since } W_1, W_2, \ldots, W_n \text{ are } i.i.d\text{)} \\ &= 1 - [1 - F_W(t)]^n \\ &= 1 - t^{-n} \end{aligned}$$

$$\therefore\ f_T(t) = F_T'(t) = \frac{d}{dt}(1 - t^{-n}) = nt^{-n-1}$$

That is,

$$f_T(t) = \begin{cases} nt^{-n-1} & t > 1, \\ 0 & \text{otherwise.} \end{cases}$$

[3 marks]

iii. From part b(ii), the density of T_1 given by

$$f_{T_1}(t) = \begin{cases} 3t^{-4} & \text{if } t > 1, \\ 0 & \text{otherwise.} \end{cases}$$

So,

$$\begin{aligned} P(T_1 > 2) = \int_2^\infty 3t^{-4}\,dt &= \left[-t^{-3}\right]_2^\infty \\ &= 0 - (-\frac{1}{8}) = \frac{1}{8} \end{aligned}$$

[2 marks]

(b) i. Let T represent the time until a major engine failure in a new car, where $T \sim Exp(\frac{1}{8})$.

$$P(T > 3) = \int_3^\infty \frac{1}{8} e^{-\frac{1}{8}t}\, dt \quad = \quad e^{-\frac{3}{8}} \approx 0.68729$$

[2 marks]

ii. Given Y represent the cost of the first major repair of a new car's engine.

$$\begin{aligned} E(Y) &= 0 \cdot P(0 \leq T \leq 3) + 8000 \cdot P(T > 3) \\ &= 0 + 8000 \cdot (e^{-\frac{3}{8}}) \\ &\approx \$5498.31 \end{aligned}$$

[2 marks]

[Total 11 Marks]

4. *All parts similar to seen examples.*
CS1 syllabus objective 2.1.3

(a)

$$ImX = \{x \in \mathbb{R} : F_X \text{ is increasing at } x\} = [0,\ 3]$$

[1 mark]

(b) X does not have the probability mass function because F_X is not a step function. [1 mark]

(c) i.

$$\begin{aligned} P(X > 1) &= 1 - P(X \leq 1) \\ &= 1 - F_X(1) \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

[1 mark]

ii.

$$\begin{aligned} P(1 < X \leq 2) &= F_X(2) - F_X(1) \\ &= \frac{1}{4}(1+2) - \frac{1}{2}(1) \\ &= \frac{3}{4} - \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

[1 mark]

iii.

$$P(X > 1|X < 2) = \frac{P(1 < X < 2)}{F_X(2)}$$
$$= \frac{1/4}{3/4} \qquad \text{(Use result from part b(ii), } P(1 < X < 2) = \frac{1}{4}\text{)}$$
$$= \frac{1}{3}$$

[1 mark]

(d) From F_X, the probability density function f_X of X can be derived as:

$$f_X(x) = \frac{dF_X(x)}{dx} = \begin{cases} \frac{1}{2} & 0 \leq x \leqslant 1 \\ \frac{1}{4} & 1 < x \leqslant 3 \\ 0 & \text{otherwise} \end{cases}$$

[1 mark]

(e)

$$E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x)dx$$
$$= \int_0^1 x \cdot \frac{1}{2}dx + \int_1^3 x \cdot \frac{1}{4}dx \quad = \left[\frac{x^2}{4}\right]_0^1 + \left[\frac{x^2}{8}\right]_1^3$$
$$= \left[\frac{1}{4}\right] + \left[\frac{9}{8} - \frac{1}{8}\right]_1^3$$
$$= \frac{1}{4} + 1$$
$$= \frac{5}{4}$$

[2 marks]

[Total 8 Marks]

5. *All parts similar to seen examples.*
CS1 syllabus objectives 2.2.1, 2.2.3, 2.3.1

(a) Since $ImX = (0, 1)$. For $x \leq 0$, and $x \geq 1$, $f_X(x) = 0$. For $0 < x < 1$, the probability density function $f_X(x)$ of X is

$$\begin{aligned} f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy &= \int_0^x f_{X,Y}(x, y)dy \\ &= 2\int_0^x (x + y)dy \\ &= 2\left[xy + \frac{y^2}{2}\right]_0^x \\ &= 2\left(x^2 + \frac{x^2}{2}\right) \\ &= 3x^2 \end{aligned}$$

$$f_X(x) = \begin{cases} 3x^2 & , 0 < x < 1, \\ 0 & , \text{otherwise.} \end{cases}$$

[2 marks]

(b)

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x \cdot f_X(x)dx. \\ &= \int_0^1 x \cdot \left(3x^2\right) dx = \int_0^1 3x^3 dx. \\ &= \left[\frac{3x^4}{4}\right]_0^1 \\ &= \frac{3}{4} \end{aligned}$$

[1 mark]

$$\begin{aligned} Var(X) &= E\left(X^2\right) - [E(X)]^2 \\ &= \frac{3}{5} - \left(\frac{3}{4}\right)^2 \\ &= \frac{3}{80} \approx 0.0375 \end{aligned}$$

[1 mark]

(c)

$$\begin{aligned}
E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_{X,Y}(x,y)dydx &= \int_0^1\int_0^x xy\cdot 2(x+y)dydx \\
&= 2\int_0^1\int_0^x \left(x^2y + xy^2\right)dydx \\
&= 2\int_0^1 \left[\frac{x^2y^2}{2} + \frac{xy^3}{3}\right]_0^x dx \\
&= 2\int_0^1 \left(\frac{x^4}{2} + \frac{x^4}{3}\right)dx \\
&= 2\int_0^1 \frac{5x^4}{6}dx = \frac{5}{3}\int_0^1 x^4dx \\
&= \frac{5}{3}\left[\frac{x^5}{5}\right]_0^1 = \frac{5}{3}\cdot\frac{1}{5} = \frac{1}{3}
\end{aligned}$$

[2 marks]

(d) The correlation between X and Y is given by:

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}}$$

$$\begin{aligned}
Cov(X,Y) &= E\left(XY\right) - E(X)E(Y) \\
&= \frac{1}{3} - \frac{3}{4}\times\frac{5}{12} \\
&= \frac{1}{3} - \frac{5}{16} \\
&= \frac{1}{48}
\end{aligned}$$

$$\therefore \rho(X,Y) = \frac{\frac{1}{48}}{\sqrt{\frac{3}{80}}\sqrt{\frac{43}{720}}} \approx 0.440225$$

[3 marks]

A correlation coefficient of 0.4402 indicates a moderately strong positive linear relationship between variables X and Y. [1 mark]

(e) For $0 < y < x < 1$, the conditional density $f_{Y|X}(y|x)$ is given by

$$\begin{aligned}
f_{Y|X}(y|x) &= \frac{f_{XY}(x,y)}{f_X(x)} \\
&= \frac{2(x+y)}{3x^2} \\
&= \frac{2}{3}\left(\frac{1}{x} + \frac{y}{x^2}\right)
\end{aligned}$$

$$f_{Y|X}(y|x) = \begin{cases} \frac{2}{3}\left(\frac{1}{x} + \frac{y}{x^2}\right) & 0 < y < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

[2 marks]

To compute $P(Y \leq \frac{1}{3}|X = \frac{1}{2})$, we first determine the conditional probability density function $f_{Y|X}(y|\frac{1}{2})$:

$$f_{Y|X}(y|\frac{1}{2}) = \frac{2}{3}\left(\frac{1}{\frac{1}{2}} + \frac{y}{\frac{1}{4}}\right) = \frac{4}{3}(1+2y), \quad 0 < y < \frac{1}{2}.$$

Next, we calculate the probability as an integral:

$$P(Y \leq \frac{1}{3}|X = \frac{1}{2}) = \int_0^{\frac{1}{3}} \frac{4}{3}(1+2y)\,dy = \frac{4}{3}\left[y + y^2\right]_0^{\frac{1}{3}} = \frac{4}{3}\left(\frac{1}{3} + \frac{1}{9}\right) = \frac{16}{27} \approx 0.59259.$$

[2 marks]

[Total 14 Marks]

6. *All parts similar to seen examples.*
CS1 syllabus objectives 2.1.1, 2.5.1

(a) To find the constant k,

$$\begin{aligned} \sum_x p_X(x) &= 1 \\ \Rightarrow \sum_{x=1}^{3} k(2x+3) &= 1 \\ \Rightarrow\ 5k + 7k + 9k &= 1 \\ \Rightarrow\ 21k &= 1 \\ k &= \frac{1}{21} \end{aligned}$$

[1 mark]

(b)

$$\begin{aligned} E(X) &= \sum_x x \cdot p_X(x) \\ &= \sum_{x=1}^{3} x \cdot \frac{(2x+3)}{21} \\ &= \frac{1}{21} \sum_{x=1}^{3} \left(2x^2 + 3x\right) \\ &= \frac{1}{21}(5 + 14 + 27) \\ &= \frac{46}{21} \approx 2.19047 \end{aligned}$$

[1 mark]

$$\begin{aligned} Var(X) &= E\left(X^2\right) - [E(X)]^2 \\ E\left(X^2\right) &= \sum_x x^2 p_X(x) \\ &= \sum_{x=1}^{3} x^2 \cdot \frac{(2x+3)}{21} \\ &= \frac{1}{21} \sum_{x=1}^{3} \left(2x^3 + 3x^2\right) \\ &= \frac{1}{21}(5 + 28 + 81) = \frac{114}{21} \\ &= \frac{38}{7} \approx 5.42857 \end{aligned}$$

$$
\begin{aligned}
Var(X) &= \frac{38}{7} - \left(\frac{46}{21}\right)^2 \\
&= \frac{38}{7} - \frac{2116}{441} \\
&= \frac{278}{441} \approx 0.63039
\end{aligned}
$$

[2 marks]

(c) For $S = \sum_{i=1}^{100} X_i$, then

$$
\begin{aligned}
E\,(S) &= 100\,\mathrm{E}(X_i) = 100(\frac{46}{21}) = \frac{4600}{21} \approx 219.048 \\
Var\,(S) &= 100\,\mathrm{Var}(X_i) = 100(\frac{278}{441}) = \frac{27800}{441} \approx 63.039
\end{aligned}
$$

[2 marks]

By Central Limits Theorem, $S \sim N(\frac{4600}{21}, \frac{27800}{441})$

$$
\begin{aligned}
P\,(210 \leqslant S \leqslant 230) &= P\,(209.5 \leqslant S \leqslant 230.5) \qquad \text{(continuity correction)} \\
&= P\left(\frac{209.5 - 219.048}{\sqrt{63.039}} \leq \frac{S - E(S)}{\sqrt{Var(S)}} \leq \frac{230.5 - 219.048}{\sqrt{63.039}}\right) \\
&\approx P(-1.20 \leqslant Z \leqslant 1.44) \qquad \text{where } Z \sim N(0,1) \\
&= \Phi(1.44) - \Phi(-1.20) \\
&= \Phi(1.44) - [1 - \Phi(1.20)] \\
&= 0.9251 - (1 - 0.8849) \qquad \text{(from the normal distribution table)} \\
&= 0.81
\end{aligned}
$$

[3 marks]

[Total 9 Marks]